Is photoperiod the dominant or least important cue for woody plant spring phenology?

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August 27, 2026

Abstract

Photoperiod is critical to plant function, entraining the circadian clock, settling the length of sunlight for photosynthesis and other stuff. But how much it matters to woody plant leafout and flowering has become less clear given more and more data. That's a bummer....

Daylength is a critical controller of growth and reproduction in plants. It defines how long they can photosynthesize each day and determines flowering success in many species. While decades of research have identified the photoperiods different species appear to need for major events, such as leafout and flowering, new research suggests gaps in our understanding of exactly how photoperiod controls fundamental plant biology CITES. At the same time, anthropogenic climate change has shifted plant events across the globe as temperatures rise—driving a growing literature of when and how photoperiod may slow this acceleration CITES.

Recent climate change has challenged our biological understanding of the dual roles of temperature and daylength to plant phenology—the timing of recurring life history events, such as leafout and flowering—with major implications for forecasting. Because forests currently sequester a large portion of fossil fuel emissions, how much they shift their growth and how successfully they can reproduce with continued warming is critical to forecasting future climate change itself CITES. Further, accurate phenological forecasts guide many forest restoration programs and crop plantings, especially of woody plants CITES. If plants mainly respond to temperature then they are more likely to shifts in step with continued warming. In contrast, if they rely on photoperiod they are more likely to lag in their responses CITES. Teasing out how much woody plants rely on photoperiod versus temperature has thus become an increasingly important topic.

Decades of work before significant anthropogenic climate change identified three major cues that plants rely on to time leafout and flowering: chilling (cool winter temperatures), forcing (warm spring temperatures) and photoperiod. Studies of model systems in annual/biennial plants show how successful flowering generally requires vernalization (chilling that leads to flowering) to make sure flowering takes place at the correct time seasonally alongside an internal circadian clock that relies on photoperiod to further determine timing CITES. These required—or 'critical photoperiods'—give rise to the terms 'short-day' and 'long-day' plants. Plants grown without vernalization or required daylengths will often flower at less optimal times climatically and/or fail to flower altogether CITES (AND reskim Wilcek). Forcing effectively drives higher rates of cellular processes and photosynthesis, producing flowing in the end, after timing determined by vernalization and circadian clock pathways CITES.

This general model is widely assumed to operate for woody plant leafout and flowering as well CITES, but has been studied far less closely. While studies in annual/biennial plants have designed experiments to separate out effects of an actual photoperiodic trigger from effects of increased photosynthesis (e.g., though night-interruption and manipulating the type and quality of light), experiments with woody plants almost never apply such methods CITES(BUT see Okie, Japanese studies). Instead, they generally focus on factorial experiments that manipulate temperature and light, generally including one shorter (e.g., 8 hour) and longer (e.g., 16 hour) photoperiod treatments with continuous high light CITES(Ailene). This approach has the benefit of directly comparing the effect of temperature versus daylength, but cannot tease out the photosynthetic effects of photoperiod that may effective drive the event to occur (similar to forcing), from the presence of 'critical photoperiods' that trigger the event.

Whether woody plants have critical photoperiods that determine leafout and flowering has major implications for forecasts in agriculture, forestry and even for climate change itself, but is still widely debated. For over two decades studies have declared that photoperiod is either the most important CITES or least important cue for woody plant spring phenology. A recent meta-analysis of experiments on woody plant leafout found the effect of photoperiod was far smaller than effects of chilling or forcing CITES, seeming to conclude the debate. Yet the study also found a persistent effect of photoperiod that appeared evolutionarily conserved across most species.

The evolutionary signal in photoperiod hints at a conserved molecular basis for woody plant daylength responses, but one that mechanistically does not easily fit well within current photoperiod models. Current studies generally observe high leafout (or flowering) with low photoperiod CITES, which does not fit with traditional circadian clock models (e.g., the internal coincidence model, CITES). Further, it seems unlikely to be solely a photosynthetic effect given how consistent it is across diverse species. It may thus potentially be derived from some other model of photoperiod response. Indeed, alongside this growing search in woody plant photoperiod responses, molecular biology has found annual/biennial plant responses that suggest missing models of photoperiod responses.

Plant biology has long recognized the importance of photoperiod in triggering critical plant events, but anthropogenic climate change has challenged ...

Leafout and flowering are some of the most important events for plants, with their timing determining the success of growth and reproduction. In the temperate zone, these events often happen in the spring, when climate poses potential risks via frost but early-activity can provide priority access to resources, including sunlight and nitrogen. Spring is also a time when humans like to monitor stuff...

References

Figures

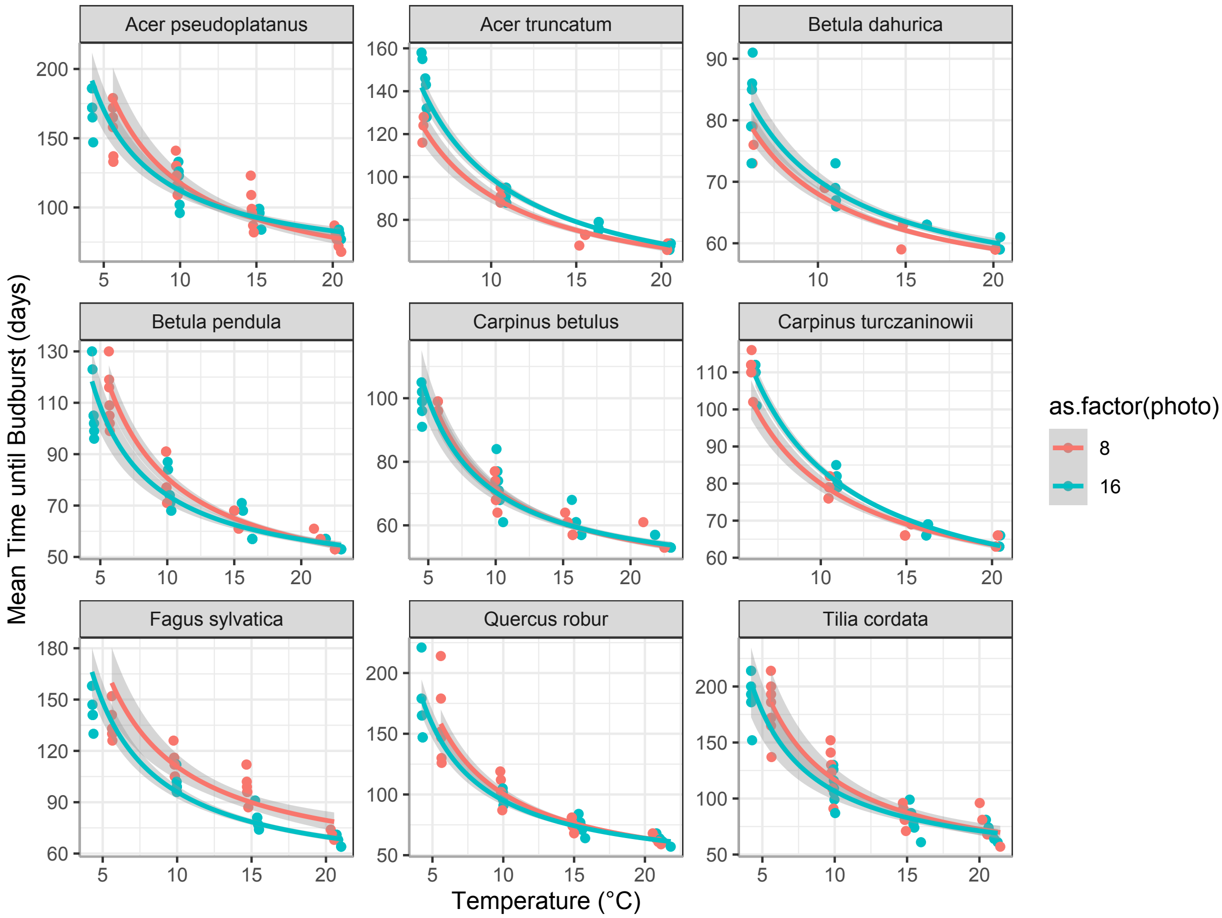

Figure 1: Results from Walde *et al.* 2023 for budburst timing across species under high chilling and two photoperiods: 8 and 16 hours. This type of study is the most common type done in woody plant phenology—it uses cuttings of dormant buds exposed to chilling (cool temperatures) followed by different combinations of warm temperatures and photoperiods (this study is unique though in covering a wide range of forcing temperatures).

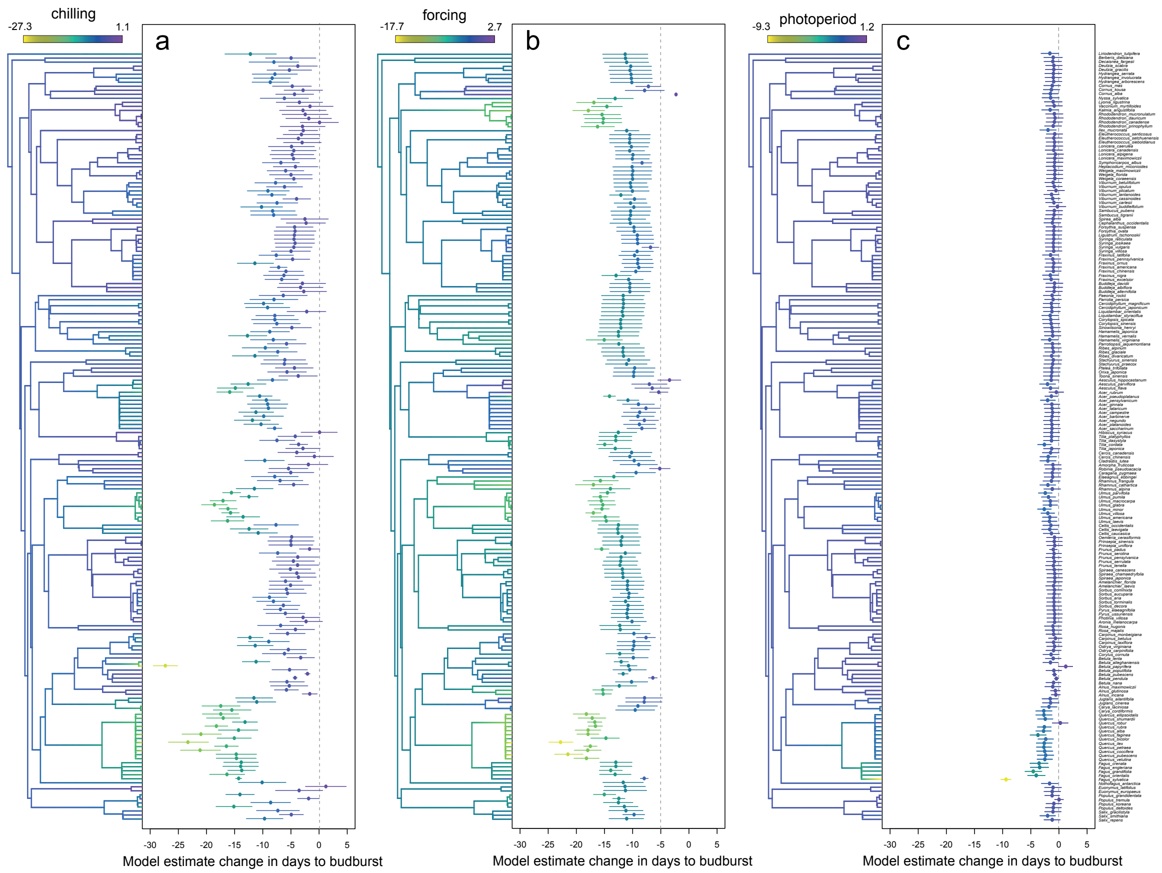

Figure 2: Estimated chilling, forcing and photoperiod responses (in units standardized by the mean and SD for each response variable) from 191 woody angiosperm species show most species shift earlier two days per standardized photoperiod unit, except for *Fagus* species with the well-studied *Fagus sylvatica* showing the strongest response. Figure reproduced from **?**.